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Solution:

Part I: $\sum_{n=-\infty}^{\infty} |h[n]| < B < \infty \implies$ BIBO-stability:

Let $x[n]$ be a bounded signal with $|x[n]| < M < \infty, \forall n$. Then:

$$\begin{aligned} |y[n]| &= |h[n] * x[n]| \\ &= \left| \sum_{k=-\infty}^{\infty} h[k] x[n-k] \right| \\ &\leq \sum_{k=-\infty}^{\infty} |h[k]| |x[n-k]| \\ &< M \sum_{k=-\infty}^{\infty} |h[k]| \\ &< MB \\ &< \infty \end{aligned}$$

Hence, we have BIBO-stability.

Part II: BIBO-stability $\implies \sum_{n=-\infty}^{\infty} |h[n]| < B < \infty$:

Since the system is BIBO-stable, for any bounded $x[n]$ ($|x[n]| < M < \infty$) we have bounded $y[n]$ ($|y[n]| < B < \infty$). Then:

$$\begin{aligned} |y[n]| &= \left| \sum_{k=-\infty}^{\infty} h[k] x[n-k] \right| < B \quad \forall n \\ \implies |y[0]| &= \left| \sum_{k=-\infty}^{\infty} h[k] x[-k] \right| < B \end{aligned}$$

Now, let $x[n] = \text{sign}(h[-n])$, which is a bounded signal since $|x[n]| \leq 1$. Then:

$$\begin{aligned} |y[0]| &= \left| \sum_{k=-\infty}^{\infty} h[k] \text{sign}(h[k]) \right| < B \\ &= \left| \sum_{k=-\infty}^{\infty} |h[k]| \right| < B \\ &= \sum_{k=-\infty}^{\infty} |h[k]| < B \end{aligned}$$

Grading: 12 points

- +6 each part

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Solution:

We will use the convolution property of the z-transform: $x[n] * h[n] \leftrightarrow X(z)H(z)$.

$$h[n] = 2^{-n}u[n] \rightarrow H(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}, \text{ ROC: } |z| > \frac{1}{2} \quad (1)$$

$$x[n] = \delta[n] - 4^{-n}u[n-1] = 2\delta[n] - 4^{-n}u[n] \rightarrow X(z) = 2 - \frac{1}{1 - \frac{1}{4}z^{-1}}, \text{ ROC: } |z| > \frac{1}{4} \quad (2)$$

$$\begin{aligned} Y(z) &= H(z)X(z) = \frac{2z}{2z-1} \left(2 - \frac{4z}{4z-1} \right) \\ &= \frac{2z}{2z-1} \left(\frac{4z-2}{4z-1} \right) \quad (\text{pole-zero cancellation}) \\ &= \frac{4z}{4z-1} \\ Y(z) &= \frac{1}{1 - \frac{1}{4}z^{-1}}, \text{ ROC: } |z| > \frac{1}{4} \\ \Rightarrow y[n] &= 4^{-n}u[n] \end{aligned}$$

At the end we have $R_Y \supset (R_X \cap R_H)$ because of the pole zero cancellation where R represents the region of convergence.

Grading: 12 points

- +8 correct method setup
- +4 correct result
- -1 for each minor error

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Solution:

(a) The transfer function can be simplified as follows:

$$H(z) = \frac{z(z-4)}{z^2-5z+6} = \frac{1-4z^{-1}}{(1-3z^{-1})(1-2z^{-1})} = \frac{-1}{1-3z^{-1}} + \frac{2}{1-2z^{-1}}, \text{ ROC: } |z| > 3$$

This ROC does not include unit circle, so it is **not BIBO stable**.

Input example: Take the inverse z-transform we can obtain the impulse response of the system:

$$h[n] = -3^n u[n] + 2^{n+1} u[n]$$

If we set $x[n] = \delta[n]$, the system output will be $h[n]$, which is unbounded.

(b) The transfer function can be rewritten as follows:

$$H(z) = \frac{z - 7}{(z + j1/3)(z - j1/3)}, \text{ ROC: } |z| > \frac{1}{3}$$

This ROC includes unit circle, so it is **BIBO stable**.

(c) The ROC is $|z| > 1$, which does not include unit circle, so it is **not BIBO stable**.

Input example: Let $x[n] = u[n]$, then $X(z) = \frac{1}{1-z^{-1}}$

$$Y(z) = X(z)H(z) = \frac{1}{1-z^{-1}} \frac{1+z^{-1}}{1-z^{-1}} = \frac{1+z^{-1}}{(1-z^{-1})^2} = \frac{1}{(1-z^{-1})^2} + \frac{z^{-1}}{(1-z^{-1})^2}$$

The corresponding $y[n] = (n+1)u[n+1] + nu[n]$ is unbounded.

(d) As $z^2 = -j = e^{j\frac{3\pi}{2}}$, we have $z = \pm e^{j\frac{3\pi}{4}}$, converting to $[-\pi, \pi]$, so $z = e^{j\frac{3\pi}{4}}$ or $e^{-j\frac{\pi}{4}}$

$$H(z) = \frac{z^{-1} - z^{-2}}{(1 - e^{j\frac{3\pi}{4}} z^{-1})(1 - e^{-j\frac{\pi}{4}} z^{-1})}, \text{ ROC: } |z| > 1$$

This ROC does not include unit circle, so it is **not BIBO stable**.

Input example: Let $x[n] = \cos(\frac{3\pi}{4}n)u[n]$, then $X(z) = \frac{1/2}{1 - e^{j3\pi/4}z^{-1}} + \frac{1/2}{1 - e^{-j3\pi/4}z^{-1}}$

$$Y(z) = X(z)H(z) = \frac{\frac{1}{2}(z^{-1} - z^{-2})}{(1 - e^{j\frac{3\pi}{4}} z^{-1})^2(1 - e^{-j\frac{\pi}{4}} z^{-1})} + \frac{\frac{1}{2}(z^{-1} - z^{-2})}{(1 - e^{j\frac{3\pi}{4}} z^{-1})(1 - e^{-j\frac{\pi}{4}} z^{-1})(1 - e^{-j\frac{3\pi}{4}} z^{-1})}$$

The first term has a double pole at $e^{j\frac{3\pi}{4}}$, after partial fractions and inverse z-transform will contain a term with a form of $ne^{j\frac{3\pi}{4}n}u[n]$ which grows unbounded.

Grading: 25 points

- (b) +4 for correct ROC and BIBO stability
- (a) (c) (d) +7 each
 - +4 for correct ROC and BIBO stability
 - +3 for correct input example (other valid input examples exist besides the example given)

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Solution:

First we find $X(z)$ and $Y(z)$, then we find the $H(z) = \frac{Y(z)}{X(z)}$. $Y(z) = \frac{1}{1-3z^{-1}} - \frac{1}{1-2z^{-1}}, |z| > 3$, and

$$X(z) = \frac{1}{1-2z^{-1}} - 6\frac{z^{-1}}{1-2z^{-1}}, |z| > 3.$$

$$\begin{aligned} H(z) &= \frac{Y(z)}{X(z)} \\ &= \frac{\frac{1}{1-3z^{-1}} - \frac{1}{1-2z^{-1}}}{\frac{1}{1-2z^{-1}} - 6\frac{z^{-1}}{1-2z^{-1}}} = \frac{(1-2z^{-1}) - (1-3z^{-1})}{(1-3z^{-1}) - 6z^{-1}(1-3z^{-1})} \\ &= \frac{z^{-1}}{1-9z^{-1}+18z^{-2}} = \frac{z^{-1}}{(1-3z^{-1})(1-6z^{-1})} \\ &= \frac{-\frac{1}{3}}{1-3z^{-1}} + \frac{\frac{1}{3}}{1-6z^{-1}} \end{aligned}$$

Since we assume the system to be causal, then the ROC is $|z| > 6$ and the unit pulse response is

$$h[n] = \frac{1}{3} (6^n u[n] - 3^n u[n])$$

The system is **not BIBO stable** because the ROC $|z| > 6$ does not contain the unit circle.

Grading: 16 points

- +8 for correct method setup
- +4 for correct result
- +4 for correct BIBO stability
- -1 each minor error

5

Solution:

a):

$$H_1(z) = \frac{2}{1-z^{-1}} - \frac{2}{1-\frac{1}{2}z^{-1}} = \frac{z^{-1}}{(1-z^{-1})(1-\frac{1}{2}z^{-1})} \quad \text{ROC: } |z| > 1$$

The ROC of $H_1(z)$ does not include unit circle, so it is **not BIBO stable**.

$$H_2(z) = 1 - \frac{\frac{3}{4}z^{-1}}{1-\frac{1}{4}z^{-1}} = \frac{1-z^{-1}}{1-\frac{1}{4}z^{-1}} \quad \text{ROC: } |z| > \frac{1}{4}$$

The ROC of $H_2(z)$ includes unit circle, so it is **BIBO stable**.

$$H(z) = H_1(z)H_2(z) = \frac{z^{-1}}{(1-\frac{1}{2}z^{-1})(1-\frac{1}{4}z^{-1})} = 4\left(\frac{1}{1-\frac{1}{2}z^{-1}} - \frac{1}{1-\frac{1}{4}z^{-1}}\right) \quad \text{ROC: } |z| > \frac{1}{2}$$

The ROC of the overall system $H(z)$ includes unit circle, so it is **BIBO stable**.

b):

Taking the inverse z-transform of $H(z)$ found in a), we have:

$$h[n] = 4 \left(\left(\frac{1}{2} \right)^n u[n] - \left(\frac{1}{4} \right)^n u[n] \right)$$

Grading: 18 points

- +3 ROC and BIBO stability of $H_1(z)$ and $H_2(z)$
- +4 correct method setup to find ROC and BIBO stability of $H(z)$
- +3 ROC and BIBO stability of $H(z)$
- +5 correct $h[n]$
- -1 each minor error

6

Solution:

a) To find the transfer function, begin by taking the Z-transform of both sides:

$$y[n] + y[n-1] + 0.25y[n-2] = x[n] + 0.5x[n-1]$$

$$Y(z) + Y(z)z^{-1} + \frac{1}{4}Y(z)z^{-2} = X(z) + \frac{1}{2}X(z)z^{-1}$$

$$Y(z)(1 + z^{-1} + \frac{1}{4}z^{-2}) = X(z)(1 + \frac{1}{2}z^{-1})$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 + \frac{1}{2}z^{-1}}{1 + z^{-1} + \frac{1}{4}z^{-2}}$$

$$H(z) = \frac{4z^2 + 2z}{4z^2 + 4z + 1}$$

$$H(z) = \frac{2z(2z + 1)}{(2z + 1)^2}$$

$$H(z) = \frac{2z}{2z + 1}$$

$$H(z) = \frac{1}{1 + \frac{1}{2}z^{-1}}, \text{ROC: } |z| > \frac{1}{2}$$

b) Using the table:

$$h[n] = \left(-\frac{1}{2} \right)^n u[n]$$

c)

$$y[n] = x[n] * h[n]$$

$$Y(z) = X(z)H(z) = \frac{1}{1 + z^{-1}} \frac{1}{1 + \frac{1}{2}z^{-1}}$$

$$Y(z) = \frac{1}{1 + \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}$$

$$Y(z) = \frac{2z^2}{2z^2 + 3z + 1}$$

$$Y(z) = \frac{2z^2}{(2z + 1)(z + 1)}$$

$$Y(z) = 1 + \frac{1}{2z + 1} + \frac{2}{z + 1}$$

$$Y(z) = 1 + \frac{1}{2} \frac{z^{-1}}{1 + \frac{1}{2}z^{-1}} + \frac{2z^{-1}}{1 + z^{-1}}$$

$$y[n] = \delta[n] + \frac{1}{2} \left(\frac{1}{2}\right)^{n-1} u[n-1] + 2(-1)^{n-1} u[n-1]$$

Grading: 17 points

- (a)
 - +4 correct method setup
 - +3 correct result and ROC
- +3 correct $h[n]$ in (b)
- (c)
 - +4 correct method setup
 - +3 correct result
- -1 each minor error